1 Processing Codes

Overview

HYDRA menu	Master data → Order → Processing codes
FEDRA menu	Detailed Scheduling → Master data → Processing codes
Transaction code	pc
Function authorization	mdpc

Purpose

Use this function to create or modify processing codes in the system.

Integration

A processing code is a collection of options that are used to control operations. Each operation references a processing code, which defines its behavior in relation to planning, collection or posting.

When an operation is created, some fields of the processing code are transferred as redundant information to the operation. Please refer to the documentation Edit operations for further information about these fields.

If you generate an order from a work plan, specific fields of the order are populated automatically. These fields are defined by the order type and were transferred to the work plan when the work plan was first created. As such, the order type is not read again when the order is generated.

You have to change the processing code and then reset the changes in order to transfer updated data from the processing code to the order or work plan.

Selection criteria

The application provides the following selection criteria:

Processing code

Unique processing code name

Field descriptions

Processing code

Processing code identification number

"SYSTEM" = default processing code that is assigned to the operation if no processing code was explicitly entered.

Comment

Designation/name or description of the processing code

Field descriptions - *General* tab

External processing OP

This option specifies whether the operation is an internal or external processing operation.

N	The operation is processed in-house.
Y	The operation is processed externally. Please note: If this option is active, the *Planning* option should be set to either "T" or "N".

Rework OP

Rework operation

Currently, this field is only used for comments and does not affect processing.

Release of rework OP

Release of the rework operation.

If this option is enabled, the initial status of the status assignment is ignored and the operation status is set to the control indicator "V" when an operation is initially created or transferred from the PPS system. Requirement: The preceding operation must be finished; i.e. it currently is assigned to a status with control indicator "E" .

Change of secondary status

The secondary status may be set for OPs with this processing code.

Can be split

V	This operation may be split. If the number of splits is transferred via the interface, then the split is performed directly in the interface.
N	The operation may not be split.

Field descriptions - *Entry* tab

Recordable

This option specifies whether operations of orders with this order type can be recorded or not, i.e. posted as defined in BDE (shop floor data collection).

Y Yes, operations can be recorded as defined in BDE.

Whether an operation can actually be recorded also depends on the setting with the same name in the *Order type*.

N The operation is for information purposes only ("info operation") and may not be posted as defined in BDE. If you attempt to log on any such operation is rejected with a validation error.

Please note:
1) To ensure that an operation does not appear in the sequencing list of the terminal, the option *sequencing list* (see below) should be disabled (set to "N").

2) An info OP is finished automatically when the subsequent OP is finished. Requirement: the operation is planned for a machine that "may be posted".

Parallel logon possible/can be logged on at the same time

☑ You can log on this operation multiple times/at the same time.

☐ You can log on this operation only once.

Sequencing list

This option specifies if operations are displayed in the sequencing list of the terminal. You can choose from the following options:

Y Yes, display in sequencing list.

Whether an operation is really displayed in the sequencing list, also depends on the setting of the same name in the *order type* and the *operation status*.

N No; orders of this order type are not displayed in the sequencing list of the terminal.

Batch management requirement

If this option is checked, batch management is required for this OP (only relevant for ADE-CHV or MPL).

Note: This information is redundantly stored for the operation. Only the information stored for the operation is relevant for processing.

Serial number obligation

If this option is checked, postings must be based on serial numbers as of this operation.

Y Yes, irrespective of whether serial numbers were transferred or not.

N No, even if serial numbers were transferred.
You can use *Y/N* to control from which OP onwards serial numbers are

required for an order.

Note: This information is redundantly stored for the operation. Only the information stored for the operation is relevant for processing.

Inspection characteristic expected This option defines whether inspection characteristics must be collected for this operation or not

X	Yes
N	No, inspection not required; no inspection characteristic expected

Underdelivery

Underdelivery is allowed for this operation. The value entered is a percentage of the target quantity. The options *Consider additionally in case of inspections* specify the values that should be integrated in the calculation apart from the yield.

Example: Enter the value 80 if the target quantity is 100 pieces and underdelivery is allowed up to a value of 80.

The system only checks the underdelivery value when the operation is logged off.

Reaction to underdelivery

	Tolerance is not checked
X	An error is generated if the tolerance quantity defined as percentage in the above-mentioned field *Underdelivery* is not reached. This validation check may not be overridden by a mandatory posting.
W	Warning if tolerance quantity is exceeded or not reached. This validation check may be overridden by a mandatory posting

Overdelivery

Overdelivery is allowed for this operation. The value entered is a percentage of the target quantity. The options *Consider additionally in case of inspections* specify the values that should be integrated in the calculation apart from the yield.

Example: Enter the value 120 if the target quantity is 100 pieces and overdelivery is allowed up to a maximum value of 120.

The system checks the overdelivery value when the operation is logged off, interrupted and when a partial quantity is confirmed/uploaded.

Reaction to overdelivery

	Tolerance is not checked
X	An error is generated if the tolerance quantity defined as percentage in the

above-mentioned field *Overdelivery* is not reached or exceeded. This validation check may not be overridden by a mandatory posting.

W Warning if tolerance quantity is exceeded or not reached. This validation check may be overridden by a mandatory posting

Notes on the options *Overdelivery* and *Underdelivery* and the corresponding *Reactions*

The options for underdelivery/overdelivery are only included in the processing code if they are not entered manually in the operation or transferred by the PPS system, and then only if a new operation is created. Reason: These values are stored redundantly in the operation. As a result, changes to the processing code have no effects on operations. Instead, the values must be changed explicitly for the operation.

But if the option *Quantity check* (see the following option) is enabled in the processing code, this option directly affects all operations with this processing code, because this information is not stored redundantly to the OP. In this case, all over/underdelivery settings that might be defined in the operation are also ignored.

Activate 100 % quantity check If this option is enabled, the **entire** target quantity of the operation must be declared as yield or scrap, i.e. the total quantity must be at least equal to the target quantity. For this reason, an operation may only be logged off if the total target quantity has been confirmed/uploaded, i.e. the total quantity has either been posted as yield or scrap.

The options *Consider additionally in case of inspections* specify the values that should be integrated in the calculation apart from the yield.

This check overrides any possibly activated validation check for under/overdelivery (see above); it cannot be overridden by a mandatory posting.

Consider additionally in case of inspections: Scrap

Include the scrap when checking the underdelivery/overdelivery fields or the quantity.

Consider additionally in case of inspections: Rework quantity

Include the rework quantity when checking the underdelivery/overdelivery fields or the quantity.

Consider additionally in case of inspections: Open quantity

Include the open quantity when checking the underdelivery/overdelivery fields or the quantity.

Field descriptions - *Posting* tab

Target quantity update

Use this option to activate a comparison of the target quantity when an operation is logged off (="FinishOP" posting).

Y	The Target quantity update is enabled for this operation, i.e. when an operation is logged off, the posted yield is transferred to the subsequent operations as target quantity. The target quantity is updated even for partial confirmations/interruptions of OPs, if the target quantity of the current operation is exceeded as a result. (is only performed online, not during recalculation!). The below-mentioned configuration "Consider additionally when terminating" is used to identify when the target quantity of the current operation is exceeded. This configuration is stored redundantly in the backlog of orders when the operation is created. The target quantity itself, however, is only compared based on the recorded yield.
N	The target quantity update is disabled for this operation. If an operation is finished, the target quantity of other operations will not be changed.

Notes

Internal business data is distributed by this function, i.e. operators of subsequent cost centers do not need to find any remaining quantities that have already accrued as scrap.

When creating an operation, this option is integrated in the operation. But you can neither display nor change it.

Quantities changed via the editing functions do not result in target quantity updates.

The function "target quantity update" does not update split operations but only the split master.

Interrupt OP instead of terminating it

This option controls how operations are logged off. Values:

☑	U = When the OP is manually logged off, it is not terminated, but instead it is simply interrupted.
☐	When the OP is manually logged off, it is terminated.

You should not use this option together with the option "Terminate preceding OPs".

If this option is used in combination with the option "Terminate OP when reaching target quantity", the option "Interrupt OP instead of terminating it" will take priority, i.e.

the OP/OPs is/are not terminated but are simply interrupted.

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Beginning of milestone

This option is relevant in connection with the option mentioned below, "Terminate preceding OPs". The processing logic is described below with the option "Terminate preceding OPs".

Please note: If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Terminate OP when reaching target quantity

An OP is automatically logged off from the workplace when the produced quantity is greater than or equal to the target quantity.

In case of operations that are logged on to several workplaces at the same time, the operation cannot be logged off automatically when the target quantity is reached. This means that the operation is only logged off from the workplace where the target quantity is reached. The operation is not logged off from other workplaces (that are logged on in parallel).

When calculating the quantity produced, the options "Consider additionally when terminating" (see below) are integrated. If none of the options is set, the quantity produced is derived from the yield only (primary quantity unit).

Possible values:

N	Do not log off automatically when reaching the target quantity (default)
Y	Log off automatically when reaching the target quantity
U	Interrupt automatically when reaching the target quantity
F	Log off automatically and log on subsequent OP with identical article number The operation that should be logged on is the operation that is planned next for this workplace (the selection criterion is the planned start of the operation). If the article number is identical to the article number of the logged off operation, this operation will be logged on to the workplace. If the article number of the operation planned next is not identical to the article number of the logged off operation, the operation will not be logged on.
K	Campaign control: Log off automatically and log on subsequent OP of the same campaign (only relevant in connection with the grouping function of HYDRA Shop Floor Scheduling, license HLS-KPG).

If the option *Terminate OP when reaching target quantity* ("Y") is enabled, an OP is interrupted when the target quantity is reached if the preceding OP*) has not yet been finished and the target quantity update is active there. As in this case, the target quantity could still increase. Otherwise, the OP is automatically logged off.

*) The option also integrates preceding OPs of other orders in case of order networks/relationships among different orders.

If this option is used in combination with the option "Interrupt OP instead of terminating it", the latter takes priority, i.e. the OP/OPs is/are not logged off, but simply interrupted.

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

If quantities are recorded automatically (e.g. via CT-UMPS), the recorded quantities are cumulated and sent at regular intervals to the server for posting. Therefore, in case of machines with rapid cycles, the operation may be logged off or interrupted with a quantity that is greater than the target quantity.

The subsequent operation is interrupted, logged off and/or logged on asynchronously/in the background. If a plausibility error occurs, the user is not informed about it. We therefore recommend to disable plausibility checks that could stop processing. An example is the plausibility check "Check running operation" in the Status assignment.

Group workplaces (workplace/machine configuration, type = G) are supported as of hymw version 7.2.1.454.

Terminate preceding OP ("milestone processing")

If an operation is logged off (posting A_AB) or terminated (posting A_BE) where the option "Terminate preceding OP" is enabled in the processing code, all directly preceding operations within the order network of the operation will be terminated.

If the preceding operations are logged off depends on their status:

- Logged on operations are logged off (posting A_AB)
- Interrupted operations are terminated (posting A_BE)

The postings result in BDE log records of record type "E".

You can control the posting using the following options:

Y All preceding OPs are automatically terminated when the OP is logged off without quantity posting (logged on OPs only if the option"External processing" is set).

M All preceding OPs are automatically terminated when the OP is logged off while the remaining quantity is automatically posted.

The options "Consider additionally when terminating" (see below) are integrated to calculate the remaining quantity. If none of the options is set, then the remaining quantity is derived from the difference of target quantity minus yield (primary quantity unit in each case).

N No automatic termination

j_ All preceding OPs are terminated automatically when the OP is logged off without quantity posting (including OPs logged on).

m_ All preceding OPs are finished automatically when the OP is logged off while the remaining quantity is automatically posted (including OPs logged on).

All preceding operations are logged off or terminated up to the operation where the option "Beginning of milestone" is enabled in the processing code. If the option is not checked, processing continues up to the first operation of the order. The processing is performed asynchronously.

If the options "J" and "M" are enabled, only those preceding OPs are finished whose current statuses are assigned to the control indicators "V" (prepared), "U" (interrupted) or "N“ (not planned) (exceptions see above).

Preceding operations that are logged on are not finished, if they are logged on to several workplaces simultaneously.

If a preceding OP is already planned for a workplace, then it is logged off with the planned workplace. If the preceding OP has only been planned for a group, the first workplace of the planned group that is not locked is identified (the workplace with the lowest number), and the OP is logged off with this workplace. If no workplace is found, the operation is finished with the workplace where the "milestone OP" was logged off.

You should not use these options together with order networks/relationships affecting several orders, as in this case not only operations of the same order are terminated, but also operations of other orders.

You should not use this option should together with the option "Interrupt OP instead of terminating it".

Operations configured with the option recordable = "N" are also terminated. In case of the options "M" and "m", operations are terminated without remaining quantity.

If you change this option subsequently in the processing code, this change has no effect on already existing operations. In the Order type , use the option "Milestone processing only for the last OP" to control whether all preceding OPs are only terminated automatically when the last OP is logged off.

Group workplaces (workplace/machine configuration, type = G) are supported as of hymw version 7.2.1.454.

Consider additionally when terminating: scrap

Scrap is also integrated when an operation is finished.

This option is integrated in both options "Terminate OP when reaching target quantity" and "Terminate preceding OP".

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Consider additionally when terminating: Rework quantity

The rework quantity is also integrated when an operation is finished.

This option is integrated in both options "Terminate OP when reaching target quantity" and "Terminate preceding OP".

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Consider additionally when terminating: Open quantity

The open quantity is also integrated when an operation is finished

This option is integrated in both options "Terminate OP when reaching target quantity" and "Terminate preceding OP".

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Automatically interrupt preceding OP

Automatically interrupts preceding OPs until the next milestone starts.

If you change this option subsequently in the processing code, this change has no effect on already existing operations.

Automatic logon by preceding OP

When an OP is logged on, the system automatically logs on all directly succeeding OPs to the planned workplace if this option is set for the OPs.

If the operation has not yet been planned for a workplace, the operation is logged on to the first workplace of the group (workplace with the lowest number).

Please note: If you change this option subsequently in the processing code, this change has no effect on already existing operations.

→**Check status of preceding OP**

This option activates a check that specifies whether or when the current operation may be logged on.

S Check status of preceding operation.
In this case, the status of the preceding operation must include the option "Successor can be logged on".

N No check.

This option only takes effect, if the option of the same name in the "Order types" is checked.

The parameter "Check status of preceding OP" is only available if the extension PCOptionsStatusPrecedingOP is activated.

Field descriptions - *Planning* tab

Synchronization with successor OP

☑ The start date of the successor OP is synchronized with the finish date of the operation. The period between these two times may not exceed the maximum synchronization time stored in the operation. When planning changes are made, the relationship between the two operations remains the same or must remain the same, which might restrict the planning options.

This function can also be used to define sequences of multiple operations for synchronization.

Application scenarios:
> First part inspection at measuring machine
> NC presetting
> Painting and drying furnace

Please note: If this option is checked, a max. synchronization time greater than zero (0) must be entered so that adjacent operations will be synchronized.

☐ Synchronization is not planned, i.e. an operation can be planned an entire shift or only one hour after the end of the preceding operation. In this case as well, you cannot plan a subsequent OP before the current operation.

Planning

This setting specifies whether and how operations are relevant for detailed planning. The following settings are available:

N No planning

T Scheduling only

The operation is scheduled, i.e. the lead time is calculated based on the process times and the dates are set accordingly. (Lead time) scheduling does not integrate competitive situations.

F Scheduling and detailed planning

The operation is relevant for detailed planning. Therefore, this operation is not only included in the scheduling but also in the automatic and/or interactive planning.

Whether or which operations are actually included in scheduling or detailed planning also depends on the configuration of the same name made in the *order type* or in the *operation status configuration*.

Overlapping

HYDRA supports the following types of overlapping:

N No, no overlapping

K Optional overlapping
Here, overlapping can be used during the reduction process. Overlapping is not used if there are enough buffers and/or no scheduling bottleneck.

S Target overlapping
scheduling and planning expect overlapping. This does not mean, however, that this possibility must be used.

You can use a send ahead quantity or lead time to specify overlapping. Both parameters describe the offset between the preceding and the subsequent operation. In case of overlapping, adjacent operations are shown internally in a start-start relationship. Times that can usually be reduced, cannot be used for reduction in this case.

Overlapping between two operations occurs if this option is enabled in the processing code of the preceding operation and neither the lead time nor the send ahead quantity of the operation is zero (0).

Otherwise, adjacent operations are shown in an end-start relationship, where transition times (if configured accordingly) can be reduced.

This document illustrates how to configure overlapping.

Cannot be interrupted

If this option is checked, the detailed planning must plan the operation so that it is not interrupted by a time without shift (e.g. shift breaks).

N Not relevant, i.e. an operation can be planned to be interrupted by times when no shift is planned.

W An operation can be planned to be interrupted by times when no shift is planned. A warning is issued, however.

Field descriptions – *Quality* tab

Inspection OP

This field identifies a QM operation. This kind of operation is not uploaded to a leading system once finished. If configured accordingly, the system automatically finishes such operations, once a related inspection step is finished.

One of the following licenses must be available to enable this field: FEP-PPF, FEP-PPE, PMV-PPK, WEP-PPW, QMS-SQM, PDV-PDM.

☑ Operations assigned to this processing code are inspection operations.

These operations only have quality control purposes and do not generate semi-finished or finished products.

☐ Operations assigned to this processing code are no inspection operations.

These operations generate semi-finished or finished products.

Logon by main OP

This parameter is only relevant for operations that are linked to a main operation.

☑ The system automatically tries to log on all linked operations, once the main operation is logged on.

The system integrates the configurations of the variable workplaces when identifying the workplaces where the linked operations should be logged on.

☐ The system does not automatically log on all linked operations when the main operation is logged on.

Logoff by main OP

This parameter is only relevant for operations that are linked to a main operation.

☑ The system automatically tries to log off or interrupt all linked operations when the main operation is logged off.

☐ The system does not try to log off or interrupt all linked operations when the main operation is logged off or interrupted.

Log OP off with still unsettled inspections

☑ If the operation is logged off or interrupted, no plausibility check is performed to find out if inspections are still pending.

☐ The system checks if there are still pending inspections when the operation is logged off or interrupted. If this is the case, the logoff or interruption of the operation is refused and a corresponding note is sent.

Interrupt main OP with still unsettled inspections

This parameter is only relevant to main operations that include linked operations.

☑ If the main operation is interrupted, the system does not check whether there are pending inspections for the linked operations.

☐ If the main operation is interrupted, the system checks for all linked operations whether or not inspections are still pending. If this is the case, the interruption of the main operation is refused and a corresponding note is sent.

Log main OP off with still unsettled inspections

This parameter is only relevant to main operations that include linked operations.

☑ If the main operation is logged off, the system does not check if there are still pending inspections for the linked operations.

☐ If the main operation is logged off, the system checks for all linked operations if inspections are still pending. If this is the case, the logoff of the main operation is refused and a corresponding note is sent.